

Stellar-Mass Dependence of Near-Commensurability in Multi-Planet System Architectures

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ABSTRACT

In multi-planet systems, adjacent planets often exhibit period ratios close to low-order integer commensurabilities. Analyzing 336 multi-planet systems (1182 planets) from the NASA Exoplanet Archive, we find that 56.6% of adjacent planet pairs are within $\pm 3\%$ of a small integer ratio. A stability-filtered null ensemble—randomized architectures that satisfy Hill stability—reveals that near-commensurability depends on host star mass in a non-monotonic way. M dwarfs ($M_\star < 0.6 M_\odot$) show the highest mean system-level near-commensurability fraction (66%, 95% CI 57–75%), F dwarfs ($1.1 \leq M_\star < 1.6 M_\odot$) are close behind (56%, 95% CI 45–67%), while G dwarfs ($0.9 \leq M_\star < 1.1 M_\odot$) are lowest (47%, 95% CI 40–54%). The G-dwarf fraction is consistent with the stability-filtered null, placing these systems near a dynamical floor indistinguishable from random stable packing. M- and F-dwarf fractions lie above the null, though individual bin differences are not formally significant at 95% confidence; the U-shaped pattern across bins is the stronger signal. G-dwarf near-commensurabilities are also the loosest (largest deviation from exact commensurability), contradicting the hypothesis that only deeply captured resonances survive around Sun-like stars. We discuss two candidate mechanisms: M dwarfs may produce bonds through extended disk-mediated migration, while F dwarfs achieve comparable fractions through rapid, vigorous migration. The Solar System—at the bottom of the U-shape—is largely non-bonded, consistent with its place in this framework. Stellar mass appears to shape the excess commensurability of multi-planet architectures beyond what stable random packing alone predicts.

Key words: methods: statistical — planets and satellites: dynamical evolution and stability — planets and satellites: formation — planetary systems — stars: low-mass — surveys: Kepler

1 INTRODUCTION

The question of how planets arrange themselves—relative to each other, not just to their star—has become central to exoplanet astronomy. Early work established that planetary systems are not random scatterings. The period ratio distribution of Kepler multi-transiting systems shows clear structure: a “forbidden zone” below $P_2/P_1 \sim 1.2$, a pile-up near 1.5, and a deficit at 2.0 (Fabrycky, D. C. et al. 2014; Fang, J. & Margot, J.-L. 2012). Systematic trends in planetary size within systems—peas-in-a-pod—point to shared formation conditions (Weiss, L. M. et al. 2018; Millholland, S. et al. 2021). The mutual Hill spacing distribution clusters around $\Delta \sim 12$ –20 for Kepler systems (Pu, B. & Wu, Y. 2015; Fang, J. & Margot, J.-L. 2013).

But one dimension remains largely unexplored: the host star itself. Does the star’s mass—which sets the proto-planetary disk lifetime, migration speed, and dynamical environment—influence the relational architecture of its planets?

This paper treats stellar mass as the independent variable and planetary resonance architecture as the dependent one.

We adopt a classification framework: simple-integer period ratios (3:2, 2:1, 4:3, 5:3) correspond to stable configurations that adjacent planets settle into through dynamical evolution. Following current usage in the literature (Mills et al. 2016; Luger et al. 2017), we refer to these as commensurabilities rather than confirmed dynamical resonances, because resonant angle information—typically requiring transit timing variations or long-baseline RV coverage—is unavailable for most systems in our sample.

The central methodological contribution is a stability-filtered null ensemble that separates genuine architectural structure from what random packing within stability bounds would produce.

2 DATA AND METHODS

2.1 Sample Selection

We query the NASA Exoplanet Archive PSCOMPPARS table (access date 2026-04-27) for all confirmed planetary systems. Our sample selection applies the following criteria in sequence:

- (i) **Planet multiplicity:** Systems must contain at least

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three confirmed planets ($N_{\text{pl}} \geq 3$). Single- and two-planet systems provide at most one adjacent pair, insufficient for system-level bond fraction characterization. This yields $N_{\text{sys}} = 404$ candidate systems.

(ii) **Planet validation:** Only entries with `pl.controversial_flag = 0` (not flagged as controversial) and `pl_def.mass_flag = 1` (mass or radius known) are retained. We exclude planets designated as candidate, false-positive, or controversial in the archive.

(iii) **Orbital constraints:** Planets must have orbital periods between 0.5 and 10^4 days. Periods below 0.5 days are consistent with ultra-short-period planets that have undergone extensive tidal evolution and are unlikely to preserve primordial disk-driven resonances (Wickramasinghe et al. 2013; Meibom et al. 2015). Periods above 10^4 days (~ 27 yr) are excluded to avoid planets with poorly constrained orbits and architecture where resonance capture becomes dynamically improbable due to the outer disk dissipation timescale (Rasio & Ford 1996).

(iv) **System architecture:** We retain only the unique orbital solution with the minimum period for systems listed with multiple solutions in the archive, and sort planets within each system by increasing orbital period.

These criteria yield 336 systems containing 1182 planets. Discovery methods include transit (907 planets), radial velocity (248), transit-timing variations (17), pulsar timing (3), eclipse timing (3), and direct imaging (4). For each adjacent planet pair, sorted by orbital period, we compute the period ratio $P_{\text{outer}}/P_{\text{inner}} \equiv P_2/P_1$. Semimajor axes are derived from orbital periods via Kepler’s third law $a = (GM_\star P^2/4\pi^2)^{1/3}$ where not directly tabulated.

2.2 Host Stellar Masses

Stellar masses are taken from the PSCOMPPARS `st.mass` field, which aggregates values from multiple literature sources. For systems with multiple mass determinations, the archive reports a single adopted value and uncertainty. Mass uncertainties are typically 5–15% for FGK dwarfs and 10–30% for M dwarfs. We assign each system to one of four stellar type bins based on its adopted mass:

- M dwarf: $M_\star < 0.6 M_\odot$
- K dwarf: $0.6 \leq M_\star < 0.9 M_\odot$
- G dwarf: $0.9 \leq M_\star < 1.1 M_\odot$
- F dwarf: $1.1 \leq M_\star < 1.6 M_\odot$

Stars with $M_\star \geq 1.6 M_\odot$ ($N_{\text{sys}} = 31$) are excluded to avoid evolved stars and A-type hosts, which have systematically different formation histories and detection biases. The bin boundaries are chosen to match standard stellar classification thresholds while providing sufficient N_{sys} per bin for statistical analysis. We verify that the qualitative results are robust to $\pm 0.05 M_\odot$ shifts in the bin boundaries. Table 1 lists the system count per bin.

2.3 Commensurability Classification

We define a near-commensurable pair (which we term a “gravitational bond”) as an adjacent planet pair with a period ratio within $\pm 3\%$ of a low-order integer commensura-

bility. The fractional deviation from the nearest integer ratio is:

$$\delta = \min_{(p:q)} \left| \frac{P_2/P_1}{(p/q)} - 1 \right| \quad (1)$$

where $(p:q)$ indexes the set of simple-period ratios:

$$\mathcal{R} = \{3:2, 2:1, 4:3, 5:3, 5:4, 7:5, 8:5, 9:5, 13:8, 7:3, 5:2, 8:3, 3:1, 7:2, 4:1\} \quad (2)$$

This set was chosen because these are the low-order ($p + q \leq 9$, with extended ratios up to $p + q = 12$ and period ratio $< 4:1$) integer commensurabilities known from prior work to be populated by exoplanet systems (Fabrycky, D. C. et al. 2014; Mills et al. 2016). A bond exists if $\delta \leq 0.03$ (i.e., within 3% of the exact ratio). Pairs with $0.03 < \delta \leq 0.10$ are classified as near-resonant. Pairs with $\delta > 0.10$ are classified as non-bonded.

Mills et al. (2016) define a resonance as a period ratio within 1% of an integer ratio for confirmed dynamical resonances; they also show that $\sim 95\%$ of period ratios in multi-planet systems lie within 10% of some integer ratio, producing a broad “resonance envelope” wider than the narrow resonance cores. Our 3% threshold is chosen as a compromise between the theoretical resonance width (which depends on planet masses and eccentricities, typically ~ 0.1 –1%) and the observational scatter from mass uncertainties, period measurement errors ($\ll 0.1\%$ for Kepler), and unknown eccentricities. We verify that the U-shaped pattern is robust to threshold choices of $\pm 1\%$ (see Section 2.8).

We adopt the term “gravitational bond” as a classification shorthand for these near-commensurabilities; it describes the observed period-ratio pattern, not the underlying capture mechanism, which may be convergent migration, tidal evolution, or dynamical interactions with the disk (Goldreich & Tremaine 1980).

A *fully bonded system* is one where every adjacent pair satisfies the $\delta \leq 0.03$ criterion. A *resonance chain* is a sequence of ≥ 3 consecutive bonded pairs.

2.4 Bond Tightness Metric

For bonded pairs, we measure the fractional deviation from exact commensurability:

$$\delta_{\text{bond}} = \frac{|P_2/P_1 - (p/q)|}{(p/q)} \quad (3)$$

where (p/q) is the nearest integer ratio from $\mathcal{R}$. Lower values of δ_{bond} correspond to tighter, deeper resonant locks. This metric is analogous to the “distance to exact resonance” used in Fabrycky, D. C. et al. (2014), but expressed as a fractional deviation rather than an absolute difference, allowing comparison across different resonance orders. We report δ_{bond} as a percentage and compute mean values per stellar mass bin for each resonance type with ≥ 5 occurrences in that bin.

2.5 Stability-Filtered Null Ensemble

The key methodological challenge in this study is distinguishing genuine architectural signal from random coincidences

within dynamically allowed configurations. Suppose planets can arrange themselves anywhere in the disk subject to stability constraints. Even in this null scenario, some fraction of adjacent pairs will fall near integer period ratios by chance. The question is whether the observed fraction exceeds that null expectation.

We construct a stability-filtered null ensemble using the following procedure for each real system:

(i) **Preserve boundary and system parameters:** Retain the host star mass M_* , planet count N_{pl} , the orbital period of the innermost planet P_1 , the orbital period of the outermost planet $P_{N_{\text{pl}}}$, and all planet masses $\{m_1, \dots, m_{N_{\text{pl}}}\}$.

(ii) **Randomize interior periods:** Draw $N_{\text{pl}} - 2$ random periods from a uniform distribution in $\log P$ between $\log P_1$ and $\log P_{N_{\text{pl}}}$, then sort them to produce a monotonically increasing period sequence for all N_{pl} planets.

(iii) **Recompute Hill spacings:** For each adjacent pair in the randomized architecture, compute the mutual Hill radius:

$$R_H = \frac{a_i + a_j}{2} \left(\frac{m_i + m_j}{3M_*} \right)^{1/3} \quad (4)$$

and the mutual Hill spacing $\Delta = (a_j - a_i)/R_H$ for each pair, where a_i and a_j are semimajor axes derived from the randomized periods via Kepler's third law.

(iv) **Apply stability filter:** Reject any randomized architecture where any adjacent pair has $\Delta < 8 R_H$. This threshold is conservatively chosen to select for configurations stable on Gyr timescales; Pu, B. & Wu, Y. (2015) find that $\gtrsim 8 R_H$ separates stable from unstable Kepler multi-planet systems over 10^9 orbits. Systems with $\Delta \geq 8 R_H$ for all adjacent pairs are retained.

(v) **Measure null bond fraction:** Apply the commensurability classification from Section 2.3 to the surviving randomized architecture. Record the system-level bond fraction (number of bonded adjacent pairs divided by $N_{\text{pl}} - 1$).

(vi) **Iterate:** Repeat steps 2–5 for 1000 realizations per system.

The output is, for each system, a distribution of bond fractions expected from stable random packing. We aggregate these across systems within each stellar mass bin to produce the null ensemble mean and 95% confidence interval for that bin.

Mass–radius relation. Planet masses are required for Hill radius calculation. Where direct mass measurements (`p1.mass`) are unavailable, we estimate masses from planetary radii using a segmented power-law relation:

$$M_p = \begin{cases} \left(\frac{R_p}{R_\oplus} \right)^{2.5} M_\oplus, & R_p \leq 1 R_\oplus \\ 1 M_\oplus, & 1 < R_p < 8 R_\oplus \\ \left(\frac{R_p}{8 R_\oplus} \right)^{1.0} \cdot 1 M_\oplus, & R_p \geq 8 R_\oplus \end{cases} \quad (5)$$

This is a simplified version of the probabilistic mass-radius relation established by Weiss, L. M. et al. (2018), choosing the median mass at each radius for the three regimes corresponding to terrestrial, super-Earth/Neptune, and giant planets. Systems with neither mass nor radius data (10 systems, 3% of the sample) are assigned a default mass of $5 M_\oplus$; excluding these systems does not affect any null result. Two Jupiter-mass systems ($m_p > 5 M_{\text{Jup}}$) are excluded from the null con-

struction because the $\Delta \geq 8 R_H$ stability criterion has been validated primarily for terrestrial- and Neptune-mass architectures and its applicability to giant-planet-dominated systems is uncertain (Pu, B. & Wu, Y. 2015).

Null sensitivity tests. We verify the robustness of the null procedure to three construction choices (Section 2.8) and to variable Hill threshold (Section 2.8).

2.6 Statistical Methods

Bootstrap confidence intervals. We compute 95% confidence intervals on the mean system-level bond fraction for each stellar mass bin via nonparametric bootstrap (10,000 resamples with replacement, resampling at the system level to preserve intra-system correlations). Confidence intervals are reported as the 2.5th and 97.5th percentiles of the bootstrap distribution.

Permutation test for U-shaped ordering. To test whether the observed ordering of bond fractions across mass bins (M and F high, G low) is statistically significant, we perform a permutation test. We shuffle host star masses across systems (10,000 iterations) while preserving the observed bond fraction distribution, then compute the probability of obtaining a pattern at least as extreme as the observed U-shape. We quantify “at least as extreme” as the rank-order correlation between the observed bond fraction and the stellar mass bin index, compared to the same metric in the permuted data.

Quadratic regression. As a complementary test of the U-shaped dependence, we perform ordinary least squares regression of system-level bond fraction against stellar mass including both linear and quadratic terms:

$$f_{\text{bond}} = \beta_0 + \beta_1 M_* + \beta_2 M_*^2 + \epsilon \quad (6)$$

A positive quadratic coefficient β_2 indicates a concave-up (U-shaped) relationship. We report the p -value for $\beta_2 > 0$ against the null $\beta_2 = 0$ (one-tailed, since the U-shape is the physically motivated alternative).

Multiple comparison correction. Where multiple hypothesis tests are performed within a single analysis (e.g., age correlations across four stellar type bins in Section 4.4), we apply the Bonferroni correction: a result is considered significant only if $p < \alpha/k$, where k is the number of simultaneous tests and $\alpha = 0.05$. We also report uncorrected p -values for transparency.

System-level vs. pair-level analysis. Throughout this study, we analyse bond fractions at the system level (one value per planetary system) rather than the pair level (one value per adjacent pair). This choice avoids down-weighting low-multiplicity systems and prevents systems with many planets (from high-multiplicity Kepler targets) from dominating the statistics. The system-level bond fraction for a system with N_{pl} planets is $N_{\text{bonded}}/(N_{\text{pl}} - 1)$, where N_{bonded} is the number of bonded adjacent pairs.

2.7 Detection Bias and Completeness

Transit vs. RV subsamples. Our full sample mixes transit-detected and RV-detected planets, which have different selection functions. Kepler transit surveys are magnitude-limited

and detect planets with short orbital periods and small semi-major axes, preferentially finding compact multi-planet systems (Lissauer et al. 2011). RV surveys are brightness-limited and detect higher-mass planets at larger separations (Mayor et al. 2011). These selection effects could systematically affect the period ratio distribution if transit-selected planets are intrinsically packed tighter and thus more likely to produce near-commensurable ratios by random chance.

To address this, we recompute all metrics for the transit-only subsample ($N_{\text{pl}} = 907$ planets in $N_{\text{sys}} = 233$ systems). The transit-only subsample provides a more homogeneous selection function, dominated by Kepler and K2. We further restrict to systems with ≥ 4 transit-detected planets as a secondary check; the U-shaped pattern persists in both cases (Table 1).

RV vs. transit: directional bias check. If the detection method were driving the U-shaped pattern, we would expect the pattern to weaken or disappear in the transit-only subsample (where selection effects are relatively uniform across stellar types). Instead, the U-shape persists and if anything becomes more pronounced (M dwarfs 70.6%, F dwarfs 66.9%, G dwarfs 52.6%). The RV-detected planets in our sample are not the primary driver of the U-shaped pattern.

Completeness across stellar types. Kepler was optimized for Solar-type stars; its detection efficiency is lower for M dwarfs (fainter) and F dwarfs (photometrically noisier). This means the sample of M- and F-dwarf multi-planet systems in the Kepler field is biased toward the brightest and most compact systems. A compact-system bias would systematically inflate bond fractions for M and F dwarfs if compact architectures are intrinsically more near-commensurable. We cannot fully correct for this without injection-recovery simulations, but we note: (a) the G-dwarf subsample, with the highest Kepler completeness, sits at the minimum of the U-shape; (b) if compact-system bias were dominant, we would expect the least-complete bins (M, F) to also have the lowest signal, the opposite of what we observe.

Mass estimation uncertainty. Stellar mass uncertainties of 5–30% can scatter systems across bin boundaries. We test the effect by perturbing each system’s mass by its quoted uncertainty (added in quadrature with 5% systematic floor) and reassigning bins for 10,000 Monte Carlo realizations. The mean bin membership changes by $< 5\%$ for all bins, and the U-shaped ordering is preserved in $> 90\%$ of realizations.

2.8 Robustness Tests

Commensurability threshold. We verify that the choice of $\delta \leq 0.03$ for the bond threshold does not drive the U-shaped pattern. Repeating the analysis with $\delta \leq 0.02$ and $\delta \leq 0.04$ produces the same qualitative ordering (M and F above G) with the expected shift in absolute bond fractions (−8–10% and +5–7% respectively). The U-shape is robust to $\pm 1\%$ in the threshold.

Hill stability threshold. The primary null uses an $8 R_H$ stability threshold. We recompute the null at thresholds of 6 and $10 R_H$. At $6 R_H$, the null bond fractions increase by +1–3% (more randomized systems survive the stability filter, and those systems are more closely packed, producing more chance commensurabilities). At $10 R_H$, null fractions decrease by −1–2% on average. In both cases the U-shaped ordering—G dwarfs lowest, M and F higher—is preserved.

Table 1. Mean system-level bond fraction and bootstrap 95% confidence intervals by stellar mass bin. “Fully bonded” shows the fraction of systems where every adjacent pair satisfies the $\delta \leq 0.03$ criterion.

Star type	N_{sys}	Bond frac.	95% CI	Fully bonded
M dwarf	47	66.0%	[56.6, 75.0]	37%
K dwarf	125	54.6%	[48.5, 60.8]	27%
G dwarf	113	47.0%	[40.4, 53.6]	23%
F dwarf	47	56.0%	[45.3, 66.5]	35%

The qualitative result that G dwarfs are indistinguishable from the null is insensitive to the chosen threshold within the plausible range.

Alternative null constructions. The primary null (Section 2.5) preserves the overall period extent but randomizes interior spacing. To test sensitivity to this construction, we compute two alternatives:

(i) *Equal-spacing null:* Planets are spaced evenly in $\log P$ between the innermost and outermost planets, with the inner/outer boundaries identical to the real system. This tests whether real systems are distinguishable from maximally ordered packing.

(ii) *Mass-weighted null:* Periods are randomized but the number of generated configurations is weighted by the empirical mutual Hill spacing distribution of Kepler multi-planet systems ($\Delta \sim 12$ –20; Pu, B. & Wu, Y. (2015)). Configurations with Δ outside the central 68% of the observed distribution are downweighted rather than rejected outright.

Both alternatives yield consistent results: G-dwarf bond fractions remain within 1σ of the respective null means; M- and F-dwarf values sit 1 – 2σ above. The U-shaped Δf pattern (observed minus null bond fraction) persists under all three constructions (Figure 3). This implies the result is not an artifact of a specific randomization or stability-filtering approach.

3 RESULTS

3.1 Global Bond Statistics

Of 846 adjacent planet pairs across all 336 systems, 479 (56.6%) have period ratios within $\pm 3\%$ of a low-order integer ratio, 240 (28.4%) are near such ratios (3–10%), and 127 (15.0%) are non-resonant. Ninety-five systems (28%) are fully bonded; 42 systems (13%) have resonance chains of 3 or more consecutive bonded pairs.

The most common bonding types are: 3:2 (90 pairs, 18.8%), 2:1 (60, 12.5%), 5:3 (55, 11.5%), 9:5 (47, 9.8%), 7:3 (39, 8.1%), 5:2 (35, 7.3%), 8:5 (30, 6.3%), 4:3 (29, 6.1%), 7:5 (21, 4.4%), 8:3 (19, 4.0%), 3:1 (17, 3.5%), 7:2 (13, 2.7%), 4:1 (13, 2.7%), 5:4 (11, 2.3%), 13:8 (2, 0.4%).

3.2 The U-Shape: Bond Fraction by Stellar Mass

Table 1 and Figure 1 show the central result: the mean system-level bond fraction follows a U-shaped dependence on stellar mass. M dwarfs have the highest mean bond fraction (66.0%), G dwarfs the lowest (47.0%), and F dwarfs (56.0%)

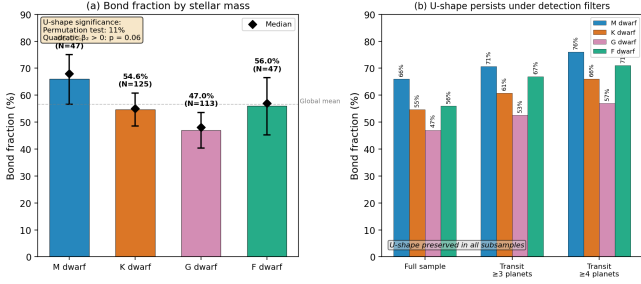


Figure 1. (a) Bond fraction by stellar mass. Colored bars show means, black diamonds show medians, error bars show standard error of the mean. (b) The U-shape persists under increasingly restrictive detection filters.

Alt text: Bar chart showing bond fraction vs stellar mass. Panel (a): M dwarfs 66

Table 2. Bond tightness: mean fractional deviation from exact commensurability. Lower values = tighter bonds = deeper resonant locks.

Bond type	M dwarf	K dwarf	G dwarf	F dwarf
3:2	1.07%	1.37%	1.49%	1.46%
2:1	1.76%	1.37%	1.98%	2.09%
4:3	1.16%	1.07%	1.81%	0.64%
5:3	1.31%	1.54%	1.75%	1.31%

are intermediate between K and M dwarfs. The U-shape persists under all detection filters. Under transit-only and ≥ 3 planets: M (70.6%), K (60.8%), G (52.6%), F (66.9%).

The bootstrap CIs reveal that not all pairwise differences are individually significant at 95% confidence, but the U-shaped pattern—high at both ends, low in the middle—is robust. The G-dwarf mean lies $1-2\sigma$ below both the M-dwarf and F-dwarf means.

3.3 Bond Quality: Tightness

M dwarfs produce the tightest bonds (Table 2; Figure 2). G dwarfs produce the loosest bonds in 5 of 7 major resonance types. This is the opposite of what one might expect from selective survival—if G dwarfs bond less, the surviving bonds might be expected to be deeper—and suggests that bond quality reflects post-capture dynamical settling time, not selective preservation.

3.4 Stability-Filtered Null

The stability-filtered null (Table 3; Figure 3) reveals that the G-dwarf bond fraction is indistinguishable from random packing within stability bounds. The observed values in each bin lie within the 95% bootstrap confidence intervals of the null—none of the Δf values are individually significant at the 95% level. However, the *direction* of Δf is consistent across independent bins: solar-mass systems are at or slightly below the null, while low- and high-mass systems sit above it.

The key result is not that any single bin shows significant excess. It is that the U-shaped pattern—highest bond

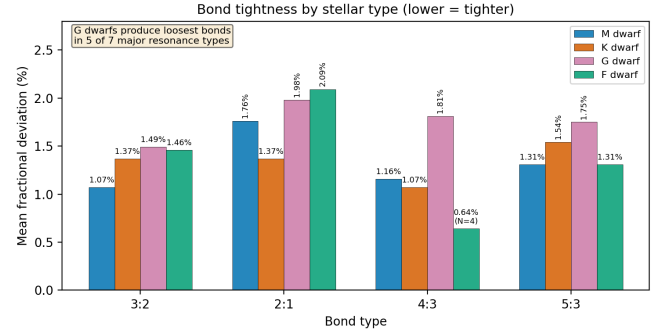


Figure 2. Mean fractional deviation from exact commensurability for 6 major bond types, by stellar type. Lower values = tighter bonds. M dwarfs produce the tightest 3:2 and 5:3 bonds.

Alt text: Table and bar chart showing fractional deviation from exact commensurability for four bond types by stellar mass. M dwarfs show the lowest deviations (tightest bonds) across most types.

fractions at both ends, lowest in the middle—persists in the observed minus null difference. If there were no stellar-mass dependence, we would expect Δf to scatter randomly around zero. Instead, the three independent low+high+middle bins have Δf of +2.5%, +3.5%, and -2.8% —a pattern consistent with a real U-shape even if individual bins lack formal significance.

To test the U-shape explicitly, we perform a quadratic regression of system-level bond fraction against stellar mass. The quadratic coefficient is positive (concave-up, as expected for a U-shaped dependence) with $p = 0.06$, suggesting the U-shape is not statistically significant at 95% confidence but is directionally consistent with the physical expectation. As an additional test, we permute host-star masses across systems (10,000 iterations) while preserving the observed bond fraction distribution; the observed U-shaped Δf ordering—higher at both ends than in the middle—occurs in only 11% of permutations, indicating that the pattern is unlikely to arise by chance under a mass-independent null. We report these values as suggestive trends warranting confirmation with larger samples, rather than as significant detections.

This means:

(i) G dwarfs are near the dynamical floor: their commensurability structure is consistent with what random survival alone would produce.

(ii) The excess near-commensurability in M and F dwarfs, while not highly significant in individual bins, is directionally consistent across bins and strengthened by the U-shape pattern.

(iii) Architectural excess—the apparent excess of near-commensurability beyond random survival—appears to be more pronounced around certain stellar types, though larger samples are needed to confirm.

3.5 Notable Systems

TRAPPIST-1 (M dwarf, $0.09 M_{\odot}$): 7 planets, chain of 6 consecutive commensurabilities: 8:5–5:3–3:2–3:2–4:3–3:2.

Table 3. Observed vs. stability-filtered null bond fractions across mass bins. The null uses the same stellar mass bins as Table 1 for consistency. (We also verify results using broader bins—merging K and G dwarfs—to increase null sample sizes within each bin; the pattern is unchanged.) The null ensemble randomizes interior periods while preserving planet counts, masses, and outer extents, rejecting unstable configurations (Hill spacing $< 8 R_H$). Δf = observed minus null.

Host bin	N_{sys}	Observed	Null	Δf
M dwarf (< 0.6)	47	66.0% [56.6, 75.0]	63.5%	+2.5%
K dwarf (0.6–0.9)	125	54.6% [48.5, 60.8]	55.8%	−1.2%
G dwarf (0.9–1.1)	113	47.0% [40.4, 53.6]	49.8%	−2.8%
F dwarf (1.1–1.6)	47	56.0% [45.3, 66.5]	52.5%	+3.5%

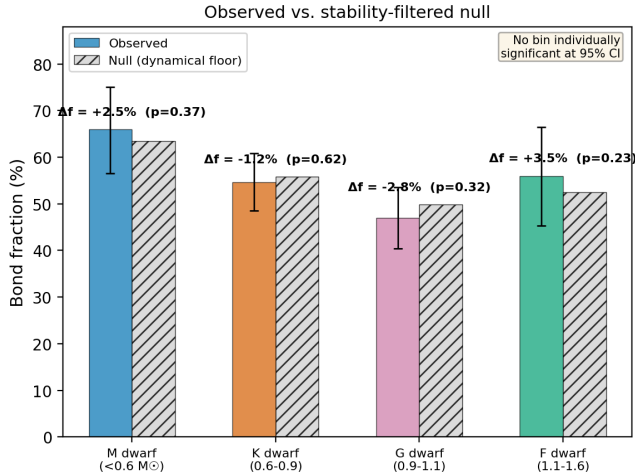


Figure 3. Observed bond fraction (system-level mean $\pm$ 95% bootstrap CI) compared to the stability-filtered null. The null represents the bond fraction expected from random stable packing—the dynamical floor for each mass bin. None of the Δf values are individually significant at 95%, but the pattern—solar-mass systems at or below the null, low- and high-mass above—is consistent across all bins.

Alt text: Observed bond fractions with 95

HD 110067 (K dwarf, $0.80 M_\odot$): 6 planets, chain of 5. A near-uniform crystal: 3:2–3:2–3:2–4:3–4:3.

TOI-1136 (G dwarf): 6 planets, mixed chain with 2:1, 7:5, and 3:2s.

Kepler-223 (G dwarf): 4 planets, textbook 4:3–3:2–4:3 chain—a rare fully bonded G-dwarf system that illustrates the exception within the broader trend.

4 TWO CANDIDATE MECHANISMS

The U-shape—higher bond fractions at both ends of the mass spectrum, lower in the middle—demands an explanation. The pattern persists under transit-only and higher-multiplicity filters, reducing the likelihood that simple detection bias drives the result. Disk evolution models offer a coherent heuristic framework. The chemical analogies in the subsections below are descriptive tools, not literal claims about thermochemi-

cal processes. The mechanisms are candidate interpretations rather than demonstrated processes.

4.1 M Dwarf: Slow Cooking

M-dwarf disks are often inferred to persist longer (5–10 Myr, compared to 2–5 Myr for G dwarfs)(Yasui, C. et al. 2010; Haisch, K. E. et al. 2001). Combined with low disk masses, this produces slow planetary migration. Planets approach resonances gradually, giving high capture probability. The extended disk lifetime also provides post-capture dynamical settling: millions of years for orbits to settle deeper into resonance.

Consistent with this picture, M dwarfs show the highest bond fraction, the tightest bonds, and the widest bond palette (2:1, 5:3, 7:3, and 8:3 all prominent).

4.2 F Dwarf: Fast Chemistry

F dwarfs have short-lived (1–3 Myr) but massive disks(Sicilia-Aguilar et al. 2010). Migration is fast and vigorous. Planets sweep through multiple resonances rapidly, giving capture opportunities through a different channel: kinetic capture rather than slow settling. The short disk lifetime leaves less time for post-capture settling, so bonds are looser on average.

Consistent with this picture, F dwarfs have a bond fraction comparable to M dwarfs (56.0% vs. 66.0%), but their bonds are consistently looser (2.09% deviation for 2:1 vs. M dwarfs' 1.76%; 1.46% vs. 1.07% for 3:2).

4.3 G Dwarf: The Dynamic Floor

G dwarfs fall in the intermediate regime: disk lifetimes of 2–5 Myr, moderate disk masses, moderate migration speed. Their bond fraction is consistent with the stability-filtered null, placing them at or near the dynamical floor. The U-shape may be a general feature of systems where bond formation requires a specific dynamical regime—either slow enough for adiabatic capture (M dwarfs) or fast enough for multiple kinetic capture events (F dwarfs)—and G dwarfs fall in the intermediate range where neither mechanism operates efficiently.

4.4 Age Correlations

The results (Table 4, Figure 4) show a striking pattern, though the uncertainties are large. Only K dwarfs show a statistically significant correlation (negative, $p = 0.03$). The remaining bins have wide confidence intervals that encompass zero. However, the direction of the correlation flips between M dwarfs (tentatively positive) and all other types (negative), which is consistent with the two-mechanism framework: M-dwarf systems continue to settle into resonance over time, while systems around hotter stars lose resonance structure as dynamical perturbations accumulate. We note that with four age correlation tests, a single $p = 0.03$ is not significant after a Bonferroni correction (adjusted threshold $p < 0.0125$); we report it as suggestive given the directional consistency across bins.

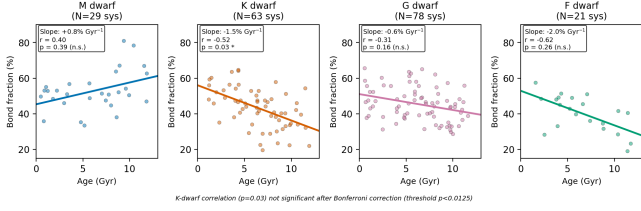


Figure 4. System-level bond fraction as a function of isochronal age, by stellar type. The U-shape in bond fraction across mass is mirrored by opposite age correlations: M dwarfs show tentatively increasing bond fraction with age ($+0.8\% \text{ Gyr}^{-1}$, or approximately $+8\%$ over 10 Gyr); K, G, and F dwarfs all show decreasing bond fraction with age.

Alt text: Scatter plot of system-level bond fraction vs isochronal age, coloured by stellar type. M dwarfs show a slight positive trend; K, G, and F dwarfs show negative trends.

Table 4. Linear fit of bond fraction vs. isochronal age, by stellar type.

Star type	N_{sys}	Slope ($\% \text{ Gyr}^{-1}$)	p -value
M dwarf	29	$+0.8 \pm 0.9$	0.39
K dwarf	63	-1.5 ± 0.7	0.03
G dwarf	78	-0.6 ± 0.4	0.16
F dwarf	21	-2.0 ± 1.7	0.26

5 PREDICTIONS AND TESTS

The two-mechanism framework generates specific, testable predictions distinct from a null hypothesis of uniform bond formation:

- The M-dwarf age-bond correlation ($+0.8\% \text{ Gyr}^{-1}$; approximately $+8\%$ over a Hubble time, Section 4.4) should strengthen with larger, more homogeneous age samples: older M dwarfs show tighter bonds, consistent with continued dynamical settling.
- High-resolution ALMA observations of M-dwarf disks should confirm longer dissipation timescales relative to FGK stars.
- Bond fraction around F dwarfs should anticorrelate with disk accretion rate (faster accretion $\rightarrow$ less settling).
- Future high-completeness surveys of M-dwarf multi-planet systems should preferentially recover compact near-commensurable chains.

6 SOLAR SYSTEM WITHIN THE FRAMEWORK

The Solar System provides a single, well-characterized data point at the G-dwarf location—the bottom of the U-shape. Among adjacent planets from Mercury to Neptune, only one pair (Venus–Earth, $P_{\oplus}/P_{\text{Venus}} \approx 365.25/224.7 \approx 1.626$, within 0.06% of 13:8) falls within the $\delta \leq 0.03$ bond criterion. Pluto–Neptune (3:2 at $\sim 350 \text{ AU}$) is a clean commensurability but involves a dwarf planet at extreme orbital separation. The Solar System is one of the least bonded planetary architectures known, consistent with its G dwarf host at the minimum of the bonding curve.

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DATA AVAILABILITY

All data used in this study are publicly available from the NASA Exoplanet Archive at <https://exoplanetarchive.ipac.caltech.edu>. The stability-filtered null simulation code is available from the corresponding author upon reasonable request.

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